

CHE655 Lab 4

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Introduction:

To understand the kth order approximations of an SVD in case of image compression

Methodology:

- The images are stored in pixel matrices in grey scale and SVD is performed.
- Finally, there is a trade-off between the memory occupied and the accuracy of the image.
- The parameter for knowing the memory storage has been taken as $k*(row+col)$ as is called memory order in the following results, where row and col are the number of rows and columns in the image matrix. The idea of taking this as parameter is because, U is row x row matrix while V is col x col matrix.

```
import numpy as np
import math
import matplotlib.pyplot as plt
import cv2

image = cv2.imread('Picture2_lab4.png', cv2.IMREAD_GRAYSCALE)

plt.imshow(image, cmap='gray')
plt.title('Selfie Image')
print(image)
plt.show()

row,col = np.shape(image)

U, sigma, Vt = np.linalg.svd(image, full_matrices=False)
U.shape, sigma.shape, Vt.shape

n = 800

reconstructed = np.matrix(U[:, :n]) * np.diag(sigma[:n]) * np.matrix(Vt[:n, :])
print(n*(row+col))
plt.imshow(reconstructed, cmap='gray')
plt.title('n = %s' % n)
plt.show()
```

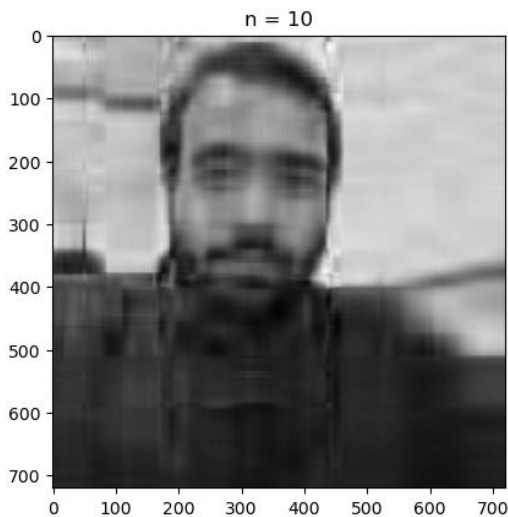
Results

Key Results:

- Selfie Image: It can be seen that optimal range for k comes between 20 and 30. At k=30 the image is quite detectable with a memory order of 43200

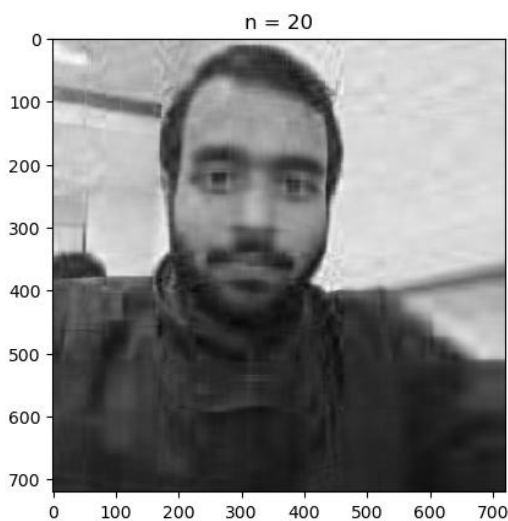
- For CT scan image (left): The optimal range turned out around $k = 400$ with memory order of 1316000
- For CT scan image (right): The optimal range turned out around $k = 700$ with memory order of 3038000
- For the CT scan, the difference is quite noticeable for the varying ranges of the k with an accuracy increasing with k .
- Even though the CT scan images have an optimal range yet it can be seen that the faithful representation needs to have a higher accuracy for better diagnosis
- In the overall analysis, it can be said that even though we must save the higher resolution image in the backend, with the low resolution compressed image acting as an abstracted version of it in the frontend memory, where space is generally less.

K=10



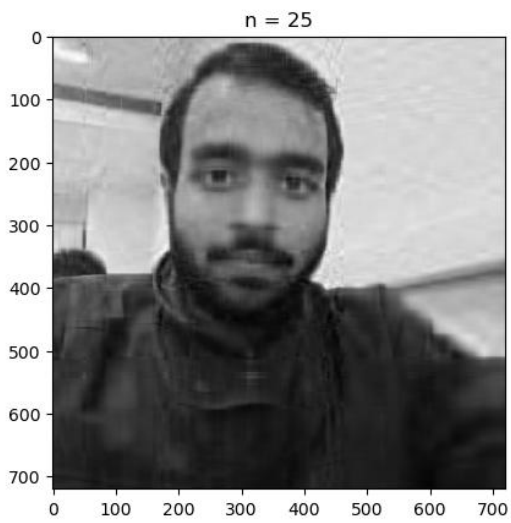
Memory order: 14400

K=20



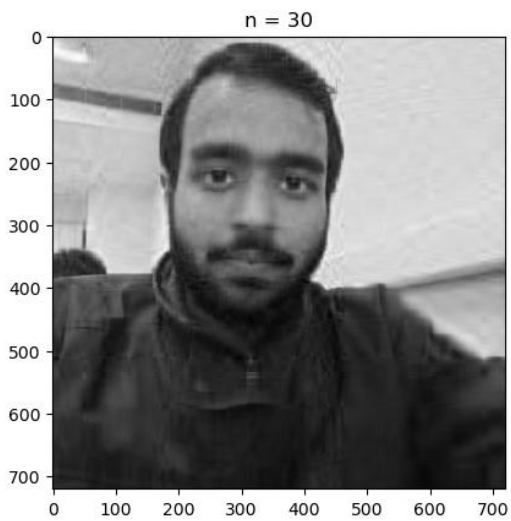
Memory order: 28800

K=25



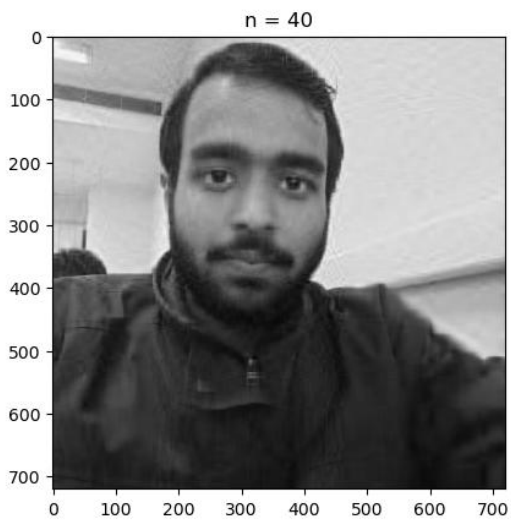
Memory order: 36000

K=30



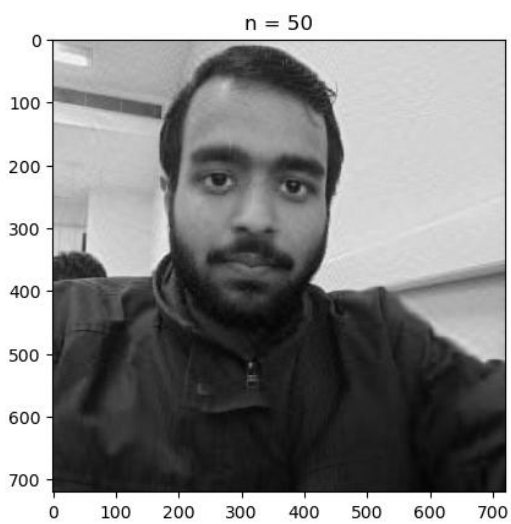
Memory order: 43200

K=40



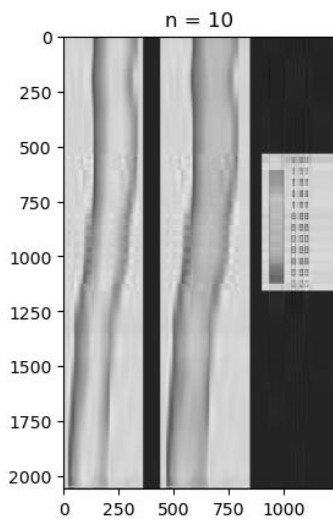
Memory order: 57600

K=50



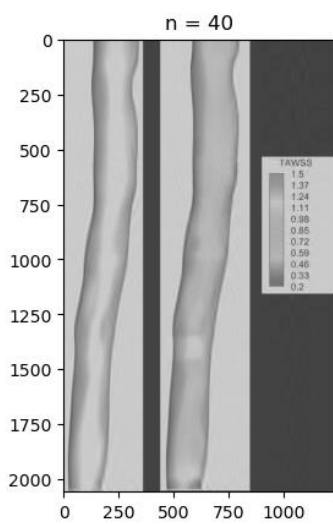
Memory order: 72000

K=10



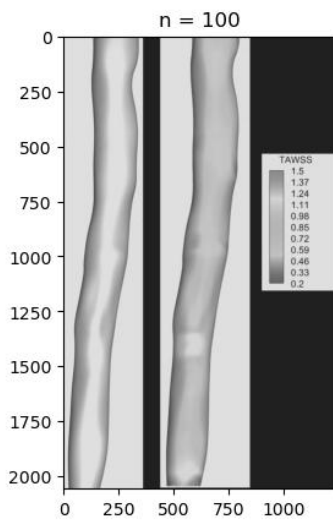
Memory order: 32900

K=40



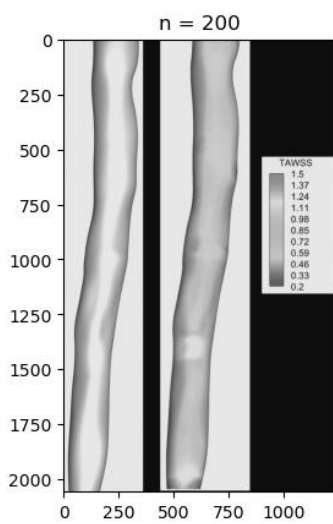
Memory order: 131600

K=100



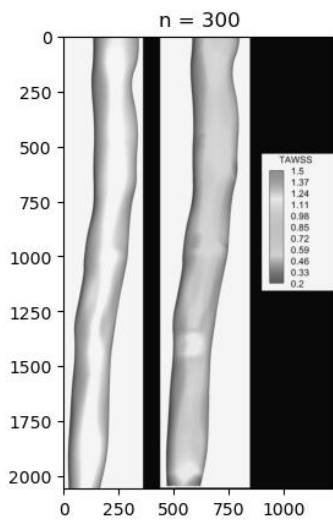
Memory order: 329000

K=200



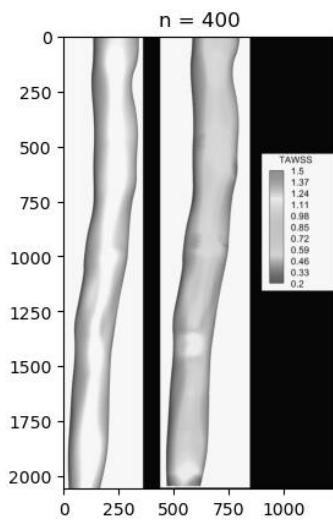
Memory order: 658000

K=300



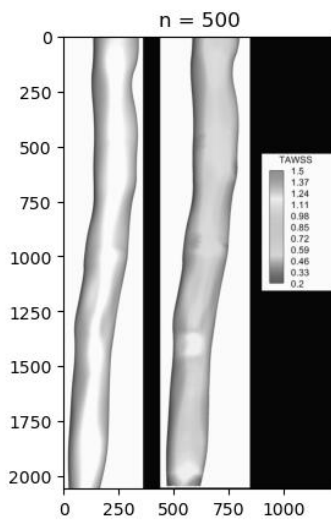
Memory order: 987000

K=400



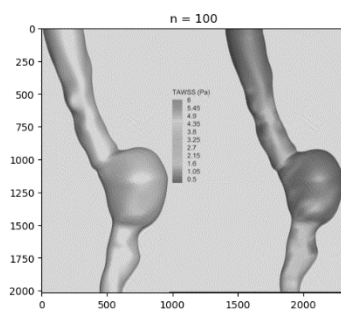
Memory order: 1316000

K=500



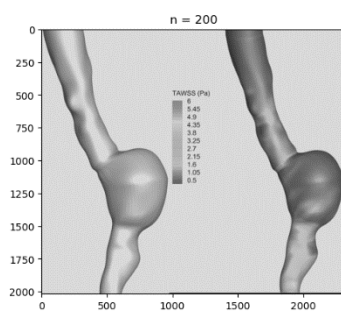
Memory order: 1645000

K=100



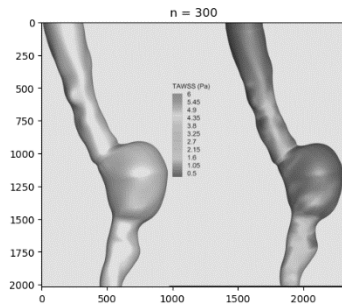
Memory order: 434000

K=200



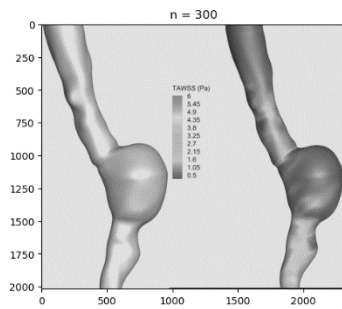
Memory order: 868000

K=300



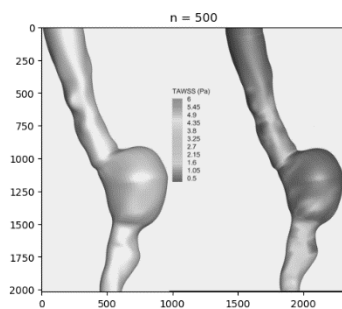
Memory order: 1302000

K=400



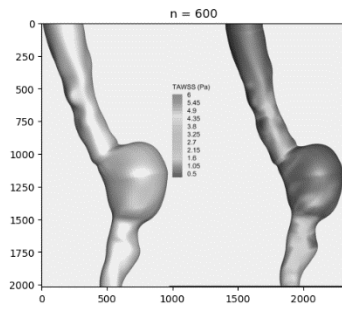
Memory order: 1736000

K=500



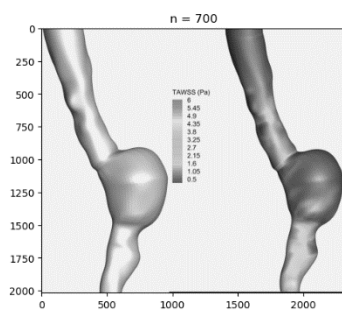
Memory order: 2170000

K=600



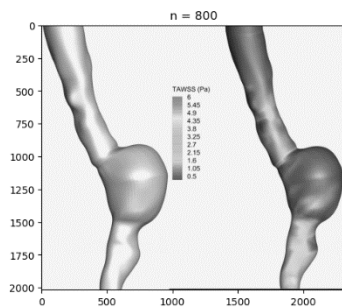
Memory order: 2604000

K=700



Memory order: 3038000

K=800



Memory order: 3472000